Mathematics practice booklet: Grade 4 expectations

Grade 4 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 4 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: 4.OA.A.1, 4.OA.A.2, 4.OA.A.3, 4.OA.B.4, 4.OA.C.5

A worked example

A ribbon is 3 times as long as a string. The string is 14 cm. How long is the ribbon? Write the equation.

1. 3 times as long means multiply: 3 x 14.
2. 3 x 10 = 30 and 3 x 4 = 12, so 30 + 12 = 42.
3. Equation: r = 3 x 14 = 42.

42 cm

Questions

1. Write as a multiplication: 24 is 4 times as many as 6.

2. Maryam has 9 stickers. Noor has 5 times as many. How many does Noor have?

3. A shirt costs AED 36 and a bag costs 3 times as much. How much do they cost together?

4. There are 47 students going on a trip and each bus holds 12. How many buses are needed?

5. List all the factor pairs of 24.

6. Is 17 prime or composite? How do you know?

7. Is 36 a multiple of 9? Explain.

8. Continue the pattern: 5, 11, 17, 23, __, __. What is the rule?

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9. A pattern starts at 2 and the rule is multiply by 2. Write the first five numbers.

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10. Ali scores 8 more points than Sami, who scores 3 times as many as Hind. Hind scores 6. How many does Ali score?

Section 2: Shapes, measuring and space

Standards practised: 4.MD.A.1, 4.MD.A.2, 4.MD.A.3, 4.MD.C.5, 4.MD.C.6, 4.MD.C.7, 4.G.A.1, 4.G.A.2, 4.G.A.3

A worked example

A rectangular football pitch for children is 40 m long and 25 m wide. Find the perimeter and the area.

1. Perimeter = 2 x (40 + 25) = 2 x 65 = 130 m.
2. Area = 40 x 25 = 1,000 square metres.

Perimeter 130 m; area 1,000 square metres

Questions

1. A rectangle has area 56 square cm and length 8 cm. What is its width?

2. Convert: 3 km = ___ m

3. Convert: 2 hours 30 minutes = ___ minutes

4. A rope is 2 m 40 cm and another is 1 m 85 cm. How much longer is the first, in cm?

5. How many degrees is a right angle? Name an angle smaller than that.

6. A turn of a clock hand from 12 to 3 turns through what angle?

7. Two angles on a straight line: one is 120 degrees. What is the other?

8. How many lines of symmetry does a rectangle that is not a square have?

9. Name a triangle that has a right angle.

10. Which pair of lines never meet however far they go: parallel or perpendicular?

Section 3: Data, graphs and chance

Standards practised: 4.MD.B.4

A worked example

A line plot shows rope lengths in quarters of a metre: 1/4 m has 3 dots, 2/4 m has 4 dots, 3/4 m has 1 dot. What is the total length of the 2/4 m ropes?

1. There are 4 dots at 2/4.
2. 4 x 2/4 = 8/4.
3. 8/4 is the same as 2 metres.

2 metres

Questions

1. Use the line plot in the example. How many ropes are there in all?

2. Which length is the most common in that line plot?

3. How much longer is the longest rope than the shortest in that line plot?

4. A line plot shows ropes in quarters of a metre: 2 dots at 1/4 m and 3 dots at 2/4 m. What is the total length of all these ropes?

5. A line plot of rainfall in centimetres, measured to the nearest half, has 3 dots at 1/2 and 2 dots at 1. What is the total?

6. A line plot shows ribbons: 2 dots at 1/4 m, 3 dots at 1/2 m and 1 dot at 3/4 m. How many ribbons are longer than 1/4 m?

7. A line plot shows cups of karak: 3 dots at 1/8 litre and 2 dots at 3/8 litre. How much karak is there in all?

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8. Survey: 12 chose karak, 9 chose juice, 6 chose water. How many more chose karak than water?

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9. In the survey above, what fraction of the 27 people chose juice? Simplify.

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10. Seven plants measure 1/2, 1/2, 3/4, 3/4, 3/4, 1, and 1 and 1/4 of a metre. How many measure 3/4 m?

Section 4: Numbers and calculating

Standards practised: 4.NBT.A.1, 4.NBT.A.2, 4.NBT.A.3, 4.NBT.B.4, 4.NBT.B.5, 4.NBT.B.6, 4.NF.A.1, 4.NF.A.2, 4.NF.B.3, 4.NF.B.4, 4.NF.C.5, 4.NF.C.6, 4.NF.C.7

A worked example

What is 3/8 + 2/8? What is 3/4 as a fraction with 8 as the denominator?

1. The denominators match, so add the numerators: 3 + 2 = 5, giving 5/8.
2. For 3/4: multiply the top and bottom by 2: 6/8.

5/8 and 6/8

Questions

1. Write in words: 45,208

2. Round 36,482 to the nearest thousand.

3. Add: 4,325 + 2,678

4. Subtract: 6,000 - 2,457

5. Multiply: 23 x 14

6. Divide: 852 / 4

7. Compare using <, > or =: 3/5 and 2/5

8. Add: 4/10 + 5/10

9. Multiply: 3 x 2/5

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10. Write 7/10 as a decimal and 0.35 as a fraction.

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11. Compare using <, > or =: 0.4 and 0.38

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12. Write 3/10 + 4/100 as a decimal.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 4 x 6 = 24
2. 45
3. AED 144 (bag AED 108, plus AED 36)
4. 4 buses (47 / 12 = 3 remainder 11, so one more bus is needed)
5. 1 x 24, 2 x 12, 3 x 8, 4 x 6
6. Prime: its only factors are 1 and 17.
7. Yes: 9 x 4 = 36.
8. 29, 35; add 6 each time.
9. 2, 4, 8, 16, 32
10. 26 (Sami 18, Ali 26)

Section 2: Shapes, measuring and space

1. 7 cm
2. 3,000 m
3. 150 minutes
4. 55 cm (240 - 185)
5. 90 degrees; an acute angle (for example 45 degrees)
6. 90 degrees
7. 60 degrees (a straight line is 180)
8. 2
9. A right triangle
10. Parallel

Section 3: Data, graphs and chance

1. 8
2. 2/4 m
3. 2/4 m (3/4 - 1/4)
4. 2 metres (2 x 1/4 = 2/4 and 3 x 2/4 = 6/4, then 2/4 + 6/4 = 8/4)
5. 3 and 1/2 cm (3 x 1/2 = 1 and 1/2, and 2 x 1 = 2)
6. 4 (3 ribbons of 1/2 m and 1 ribbon of 3/4 m)
7. 1 and 1/8 litres (3/8 + 6/8 = 9/8)
8. 6
9. 9/27 = 1/3
10. 3

Section 4: Numbers and calculating

1. Forty-five thousand, two hundred eight
2. 36,000
3. 7,003
4. 3,543
5. 322

6. 213
7. 3/5 > 2/5
8. 9/10
9. 6/5 = 1 and 1/5
10. 0.7 and 35/100 = 7/20
11. 0.4 > 0.38 (0.4 is 0.40, and 40 hundredths is more than 38 hundredths)
12. 0.34 (3/10 = 30/100, and 30/100 + 4/100 = 34/100)